

A Level Computer Science (9618)

Topical Past Paper Worksheet

Topic 2: Communication

Subtopic 2.1: Networks including the Internet

Concept 2.1.9: Bit Streaming

Total Questions: 4

Mark Schemes begin on Page 5

Question 1 | Source: 9618_s25_qp_11 (Q2.a)

- 2 Programmers in a software development company take part in live video conferences to discuss their work.

The live video conferences take place using real-time bit streaming. The video is compressed before it is transmitted.

- (a) Explain how data is transferred using real-time bit streaming.

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Question 2 | Source: 9618_w24_qp_13 (Q9.c)

- 9 A computer is connected to a Local Area Network (LAN) that connects to a Wide Area Network (WAN).

- (c) The computer on the LAN is used for real-time video conferences, where people connected to the internet communicate in real-time using video and audio.

Explain how bit streaming is used in a real-time video conference.

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Question 3 | Source: 9618_s24_qp_11 (Q2.e.i)

- 2 A video doorbell is attached to the front door of a house. The doorbell uses a motion sensor to detect when a visitor walks in front of the door. When the motion sensor is activated:
- The digital camera in the doorbell starts recording a video.
 - A message is transmitted to a smartphone so that the person who lives in the house can watch the video.

The doorbell also has a button that can be pressed. When the button is pressed, a message is transmitted to a smartphone to play the doorbell sound.

The videos are stored on the doorbell’s internal secondary storage device and overwritten when the secondary storage device is full.



(e) The video doorbell allows both real-time and on-demand bit streaming.

(i) State what is meant by bit streaming.

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Question 4 | Source: 9618_s24_qp_11 (Q2.e.ii)

- 2 A video doorbell is attached to the front door of a house. The doorbell uses a motion sensor to detect when a visitor walks in front of the door. When the motion sensor is activated:
- The digital camera in the doorbell starts recording a video.
 - A message is transmitted to a smartphone so that the person who lives in the house can watch the video.

The doorbell also has a button that can be pressed. When the button is pressed, a message is transmitted to a smartphone to play the doorbell sound.

The videos are stored on the doorbell’s internal secondary storage device and overwritten when the secondary storage device is full.



(e) The video doorbell allows both real-time and on-demand bit streaming.

(ii) Give **two** differences between real-time and on-demand bit streaming.

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MARK SCHEMES

Mark Scheme for Question 1 | Source: 9618_s25_ms_11 (Q2.a)

2(a)	1 mark each to max 4 • Video is transmitted continuously • as a series of bits • The video is uploaded to a media server • The users download from the media server • On download, the media server sends the data to a buffer on the user's device • buffer is used when there is a difference in speed between transmission and receipt • Buffer stores data from server until recipient can receive it • Recipient views bit stream from the buffer	4
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Mark Scheme for Question 2 | Source: 9618_w24_ms_13 (Q9.c)

9(c)	1 mark for each bullet point (max 4) • Data is compressed before transmitting • The video and audio are transmitted continuously as a series of bits • The video is uploaded to a media server • On download, the media server sends the data to a buffer • The buffer stores data from server until the receiving device can process it • The receiving device receives the bit stream from the buffer	4
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Mark Scheme for Question 3 | Source: 9618_s24_ms_11 (Q2.e.i)

2(e)(i)	1 mark for: Continuous ordered flow of bits over a communication path	1
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Mark Scheme for Question 4 | Source: 9618_s24_ms_11 (Q2.e.ii)

2(e)(ii)	1 mark each to max 2: • Real-time is direct from source whereas on-demand is pre-recorded/downloaded to view later • Real-time cannot be re-watched, on-demand can be paused, re-watched etc. • Real-time plays continually, on-demand downloads sections/blocks and cannot play until next section is downloaded	2
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